TASK 6: REPORT

Insurance Charges Prediction Using Linear Regression

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1. INTRODUCTION

A medical aid scheme in South Africa requires a model to predict insurance charges based on customer lifestyle and geographic information. This report details the steps taken to build and evaluate a Linear Regression model using a US-based insurance dataset as a proof of concept. The dataset was sourced from Kaggle and contains 1338 records with 7 features, including age, sex, BMI, number of children, smoking status, region, and insurance charges.

Dataset - Source: https://www.kaggle.com/datasets/mirichoi0218/insurance- Records: 1338 rows, 7 columns - Target variable: charges - Features: age, sex, bmi, children, smoker, region

Tools and Libraries

- Python 3
- Jupiter Notebook
- pandas — data manipulation
- NumPy — numerical operations
- matplotlib — data visualisation
- seaborn — statistical visualisation
- scikit-learn — machine learning model and evaluation.

2. ANALYSIS PLAN

A. Exploratory Data Analysis (EDA)

The following steps were taken during the EDA phase:

- The dataset contains 1338 rows and 7 columns with no missing values

  values, meaning no data cleaning or imputation was required

- The target variable (charges) is right-skewed, meaning most

  customers pay lower charges, while a smaller group of customers

  pay significantly higher amounts

- Smokers pay significantly higher charges than non-smokers,

  making smoking status the most influential feature in the dataset

- Age and BMI show a strong positive relationship with charges

  older patients and those with higher BMI tend to pay more

- Sex and region showed very little impact on charges

- No outliers were removed, as the high charge values belong to

  smokers and are legitimate data points that reflect

  real-world patterns.

B. Feature Selection

Categorical columns (sex, smoker, region) were encoded into numeric values using Label Encoding before training, since Linear Regression requires numeric inputs. A correlation analysis was then conducted to identify the most important features:

- Smoker had the highest correlation with charges (0.79)

- Age showed a moderate positive correlation with charges

- BMI showed a moderate positive correlation with charges

- Children had a weak but positive correlation - sex and region had very low correlations with charges. All six features were included in the initial OLS model. A second model was then trained using only the top three features (smoker, age, and BMI) for comparison purposes.

C. Model Training

The dataset was split into 80% training and 20% testing sets using train_test_split with a random state of 42 to ensure reproducibility. Four regression models were trained and compared:

1. OLS Linear Regression; baseline model
2. LASSO Regression; with automatic feature selection
3. Ridge Regression; with coefficient shrinkage
4. Elastic Net; combining LASSO and Ridge penalties

All models were trained using default hyperparameters initially. The model coefficients from the OLS model revealed the following feature importance:

- smoker: 23647.82, by far the strongest predictor
- children: 425.09, each additional child adds to charges
- BMI: 335.78, higher BMI leads to higher charges
- age: 257.06, older customers pay more
- sex: -18.79, very minimal impact
- region: -271.28, very minimal impact

```
Model Coefficients:
    Feature   Coefficient
4    smoker  23647.818096
3  children    425.091456
2       bmi    335.781491
0       age    257.056264
1       sex    -18.791457
5    region   -271.284266
```

D. Interpret and Evaluate Model

-Use the Following evaluation metrics:

-R^2 Score: measures how well the model explains the variance

-Mean Absolute Error (MAE) - average prediction error

-Mean Squared Error (MSE) - Penalises large errors

-Root Mean Squared Error (RMSE) - interpretable error in same units as charges

-Plot actual vs predicted charge to visually evaluate the model

E. Write a Report

-EDA findings and data cleaning steps

-Features selection decisions

-How the model was trained

-Evaluation metrics and what they mean

-Conclusions and recommendations for the medical aid scheme

3. Conduct Analysis

a. **Exploratory data analysis**
   Check for missing values in the dataset.
   Visualised the distribution of target variables (charges).
   Created a box plot for charges vs smoker status.
   Created a box plot for charges vs region.
   Created a box plot for charges vs sex.
   Created a scatter plot for age vs charges.
   Created a scatter plot for BMI vs charges.
   Created a correlation heatmap for numeric variables

EDA Findings

-The charge distribution is right-skewed, which means most people pay lower charges but a few of the population pay very high amounts.

-Smokers pay significantly higher charges than non-smokers

-Age and BMI show a positive relationship with the charges, that is as charges increase you wuill see higher ages and more BMI

-Region does not seem to have a strong effect on charges, as it does not directly affect the charges in any way

-There are no missing values in the dataset, as we run the missing values count we realised that they where zero missing values from the variables.

b. **Feature selection**

Encode categorical columns using label encoding.

Check the correlation of all features with charges.

Define the feature (x) and (y).

Feature Selection Findings

-Smoker has the highest correlation with charges

-Age and BMI also show stromg positive correlation with charges

-Children have weak correlation with charges

-Sex and region have very low correlations

-All features will be kept for the initial model, but smoker, age and BMI are expected to be the most important as they have a significant correlation with charges.

4. Evaluate Model

a. **Interpret and Evaluate Model**

-Predict the test set

-calculate evaluation metrics

-Determine Actual and residuals

-plot actual vs residual charges

-residuals plot

-residuals scatter plot

b. **Data Preparation and Splitting the Dataset**

This step prepares the dataset for modelling. The categorical are converted into numeric form using one-hot encoding, and the target variable (charges) is separated from the predictors. The data is then split into train and test sets for the model to be trained.

-We train the model and check residual.

The residual plot shows how the prediction errors are distributed against the predicted insurance charges. Most residuals are scattered around the zero line, which indicates strong systematic bias. This spread may increase when the predicted charge is high, which explains the model is less accurate for more expensive insurance charges. Model performs well, although some variability remains in the prediction errors.

b. **Evaluating findings**

- R^2 Score: Measures how much of the variance in charges is explained by the model. When a score is closer to 1.0 it is better.

-MAE: The average amount the model's predictions are off by in dollors.

-RSME: Similar to MAE but penalises larger errors more heavily

Interpretation

-The Smoker coefficient is largest, confirming that smoking is the strongest predictor of high insurance charges.

-Age and bmi also contribute positively to charges,

-If the R^2 score is below 0.80, the model may need retraining with different parameters.

4.1. Retrain with different Algorithms

Based on the initial Linear Regression (OLS) model, we will now retrain using three additional regularisation algorithms: LASSO, Ridge and Elastic Net, to compare performance and determine the best model for predicting insurance charges.

- LASSO Regression: adds a penalty to reduce less important feature coefficients to zero, effectively performing automatic feature selection.

LASSO Metrics:

R^2 Score: 0.7835

MAE: 4182.43

RMSE: 5797.03

-Ridge Regression: adds a penalty to shrink coefficients of less important features without removing them, works well when features are correlated with each other.

Ridge Metrics:

R^2 Score: 0.7833

MAE: 4193.59

RMSE: 5800.43

-Elastic Net: combines both LASSO and Ridge penalties, can remove unimportant features while also handling -correlated features, making it the most flexible model.

Elastic Net Metrics:

R^2 Score: 0.4186

MAE: 7423.86

RMSE: 9500.95

Model Comparison Table

Model	R^2 Score	MAE	RMSE
OLS LR	**0.7833**	**4186.51**	**5799.59**
LASSO	**0.7835**	**4182.43**	**5797.03**
Ridge	**0.7833**	**4193.59**	**5800.43**
Elastic Net	**0.4186**	**7423.86**	**9500.95**

Model Comparison Conclusion

Based on the evaluation metrics, LASSO Regression was the best performing model with:

- R^2 Score: 0.7835, explaining 78.35% of the variance in charges
- MAE: 4182.43, predictions were off by $4,182 on average
- RMSE: 5797.03, the lowest overall prediction error

OLS and LASSO performed very similarly, suggesting the dataset does not have many irrelevant features. Ridge performed marginally worse, indicating low multicollinearity between features. Elastic Net performed significantly worse with the default parameters and would require further hyperparameters to improve.

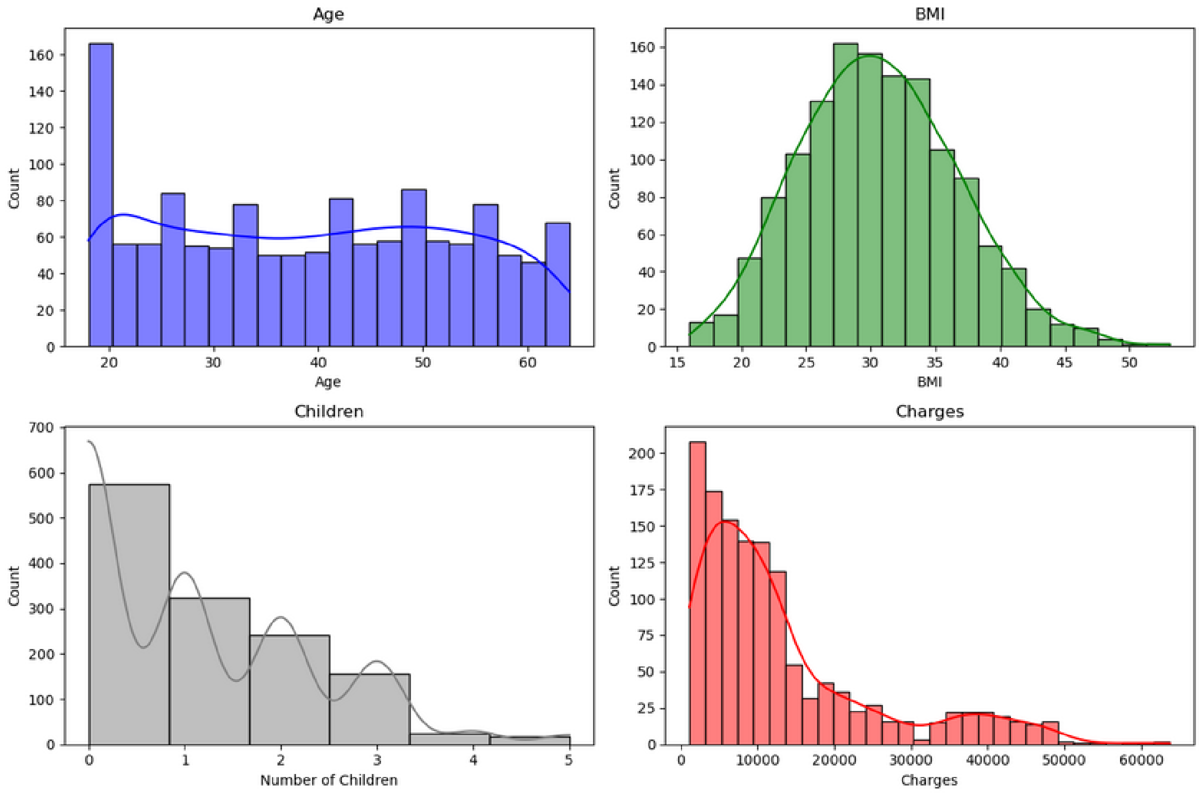
Therefore, LASSO Regression is recommended as the final model for predicting insurance charges for the medical aid scheme.

4.2. Exploratory data Analysis with other Analysis Tools

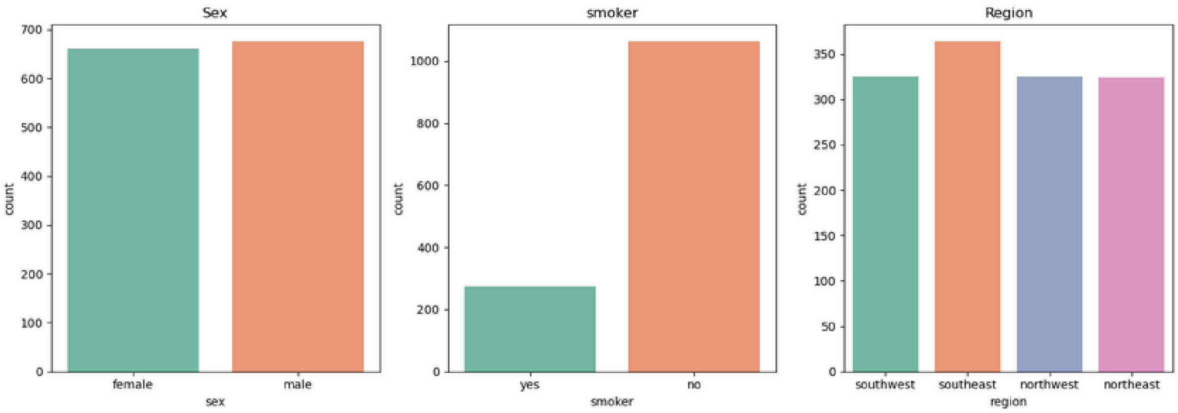
Univariate Analysis: examining each variable individually

-distribution of numeric variables
-age distribution
-BMI distribution

-children distribution
-charges distribution



- **Categorical variables**
- Sex distribution
- Smoker distribution
- Region distribution

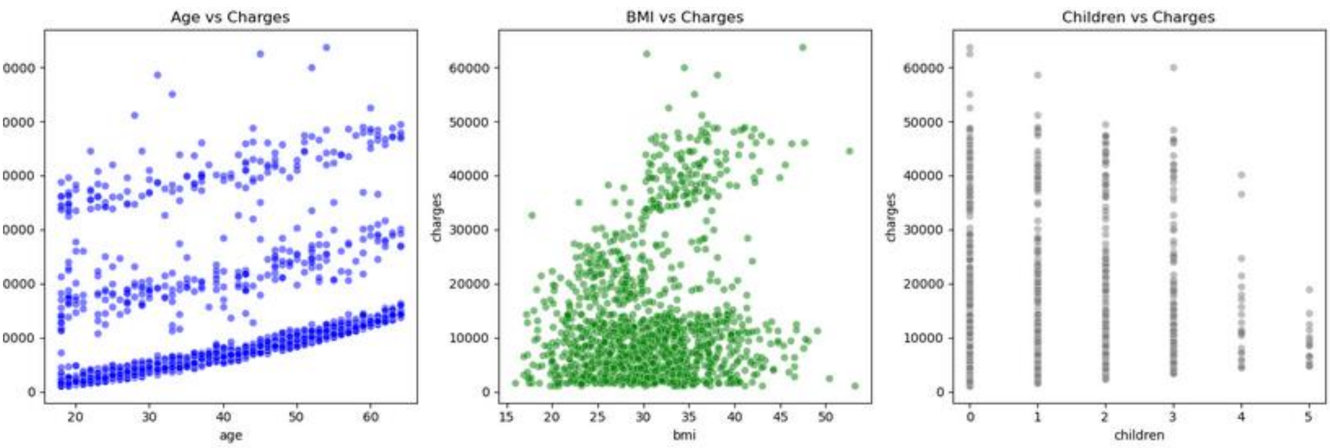


Univariate Analysis Findings
-Age is fairly uniformly distributed between 18 and 64 years

-BMI follows a roughly normal distribution centred around 30
-Most customers have 0 to 2 children
-Charges are right-skewed, most customers pay lower amounts but a small group pays significantly higher charges
-The dataset is fairly evenly split between male and female
-Non-smokers make up the majority of customers
-All four regions are fairly evenly represented

Bivariate Analysis: examining relationships between two variables.

- **Numerical features vs charges**
- Age vs charges
- Bmi vs charges
- Children vs charges



Bivariate Analysis Findings

- Age has a positive relationship with charges, older Customers tend to pay more
- BMI shows a moderate positive relationship with charges
- The number of children has a weak positive relationship with charges
- Smokers pay dramatically higher charges than non-smokers, this is the strongest relationship in the dataset

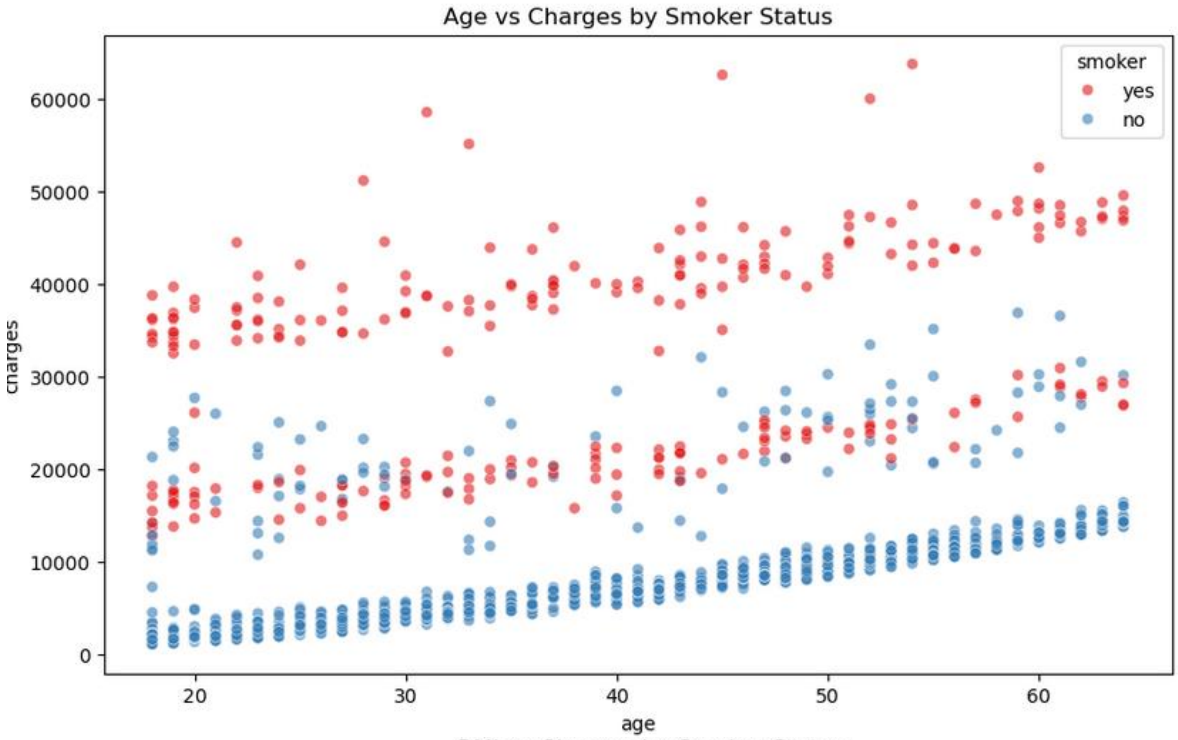
- Sex has very little impact on charges
- The region has minimal impact on charges with slight differences between regions

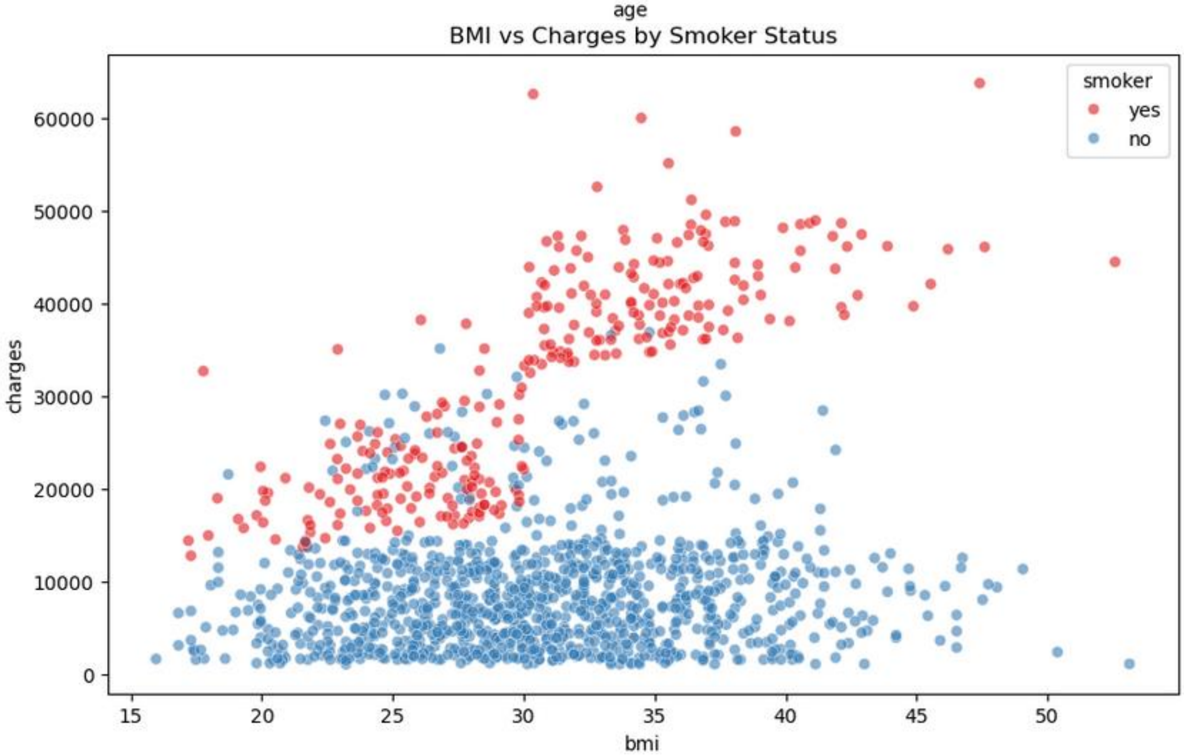
Multivariate Analysis: examining relationships between multiple variables simultaneously.

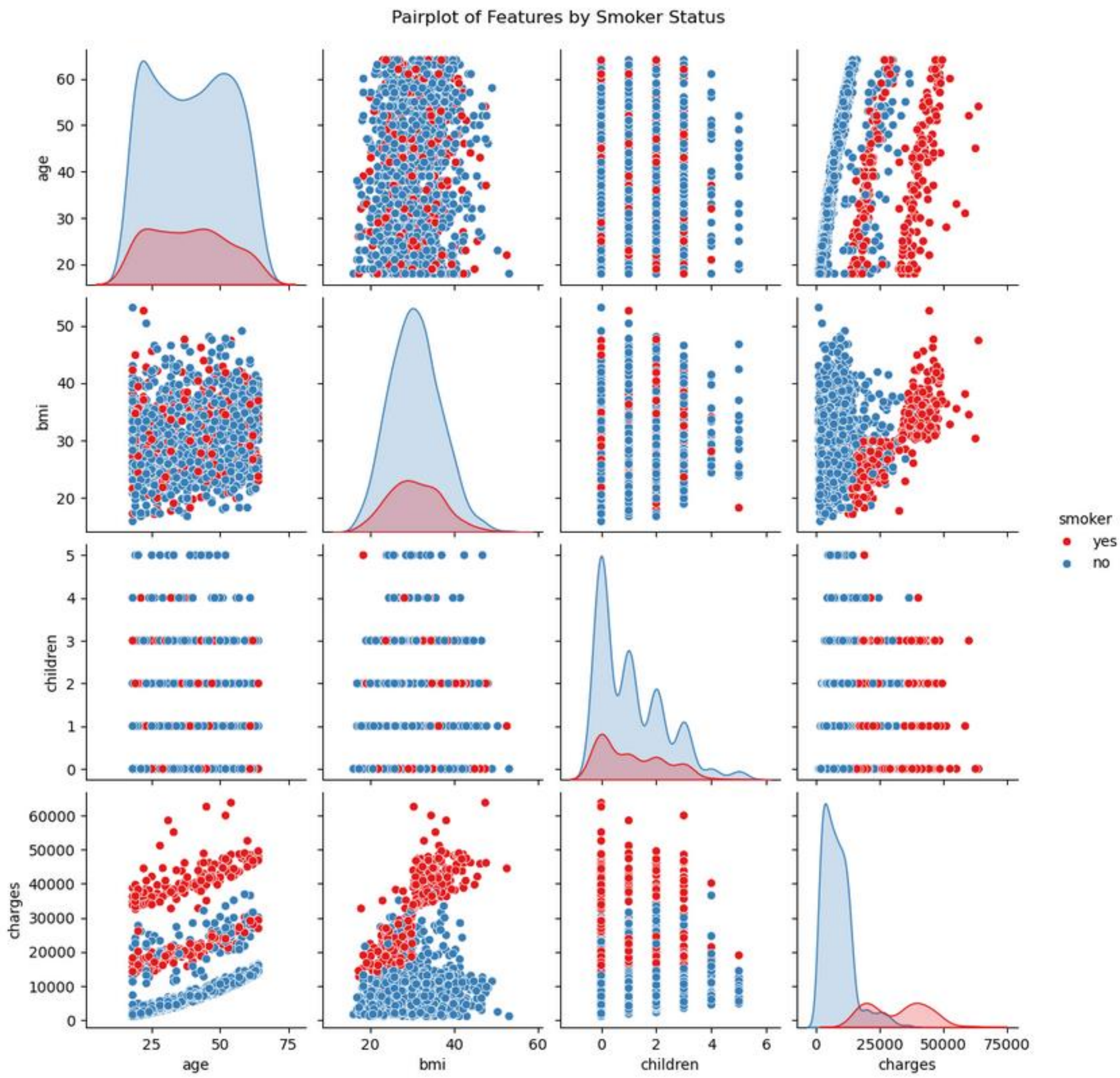
-Age vs charges coloured by smoker

-Bmi vs charges coloured by smoker

-Pairplot of all numeric variables


Age vs Charges by Smoker Status
smoker
yes
no
charges
age


BMI vs Charges by Smoker Status
smoker
yes
no
charges
bmi



Multivariate Analysis Findings

-The correlation heatmap confirms that charges have the strongest correlation with smoker status. This was plotted earlier already in our analysis, as we see under Exploratory analysis.

-Age and BMI both show moderate positive correlations with charges

-When combining age and smoker status, it is clear that older smokers pay significantly higher charges than older non-smokers

-When combining BMI and smoker status, smokers with high BMI pay the highest charges of all groups

-The pairplot confirms that smoker status creates two distinct clusters in the data, smokers consistently pay higher charges across all age and BMI ranges

-There is low multicollinearity between the input features, which is good for Linear Regression.

Task 5. Statistical Analysis

a. Null Hypothesis and P-value

Null Hypothesis (H_0): The feature has no significant relationship with insurance charges.

Alternative Hypothesis (H_1): The feature has a significant relationship with insurance charges.

A p-value below 0.05 means we reject the null hypothesis and confirm the feature is statistically significant.

P-Value Analysis Findings

Based on the statistical analysis:

-Features with p-value below 0.05 are statistically significant and should be kept in the model

-Features with p-value above 0.05 are not statistically significant and could be removed

Expected findings based on our EDA:

-smoker; expected to have very low p-value (significant)

-age; expected to have very low p-value (significant)

-Bmi; expected to have very low p-value (significant)

-Children; expected to have low p-value (significant)

-sex ; expected to have high p-value (not significant)

-region; expected to have high p-value (not significant)

Features found to be statistically insignificant ($p > 0.05$) will be considered for removal in model retraining.

Null Hypothesis Conclusion

After removing statistically insignificant features:

Model	R^2 Score	MAE	RMSE
All Features (OLS)	0.7833	4186.51	5799.59
Significant Features Only	0.7811	4213.80	5829.38

Features where we **reject the null hypothesis** ($p < 0.05$ -- statistically significant):

---age, BMI, smoker, children

Features where we **accept the null hypothesis** ($p > 0.05$ -- not statistically significant):

---sex, region

This confirms that smoking status, age, BMI, and number of children are the true drivers of insurance charges.

b. **Feature Selection**

A correlation analysis and statistical p-value analysis were conducted to identify the most important features:

Features	correlation	p-value	Significant?
Smoker	Highest	<0.05	yes
age	moderate	<0.05	Yes
bmi	moderate	<0.05	yes
Children	weak	<0.05	Yes
Sex	Very low	>0.05	no
region	Very low	>0.05	no

Features where we reject the null hypothesis ($p < 0.05$ -- statistically significant): - smoker, age, bmi, children. Features where we accept the null hypothesis ($p > 0.05$ -- not statistically significant): - sex, region.

6. Recommendations for the Medical Aid Scheme

Based on the findings, the following recommendations are made for the South African medical aid scheme:

1. Smoking status: should be the primary factor in determining insurance charges as it has the strongest influence on costs, smokers should be charged significantly higher premiums

2. Age: should be the second most important factor, a sliding scale of charges based on age is justified by the data

3. BMI: should be factored into the pricing model, customers with higher BMI should pay higher premiums

4. Number of children: has a small but statistically significant effect and should be considered in the pricing structure

5. Sex and region: have minimal impact on charges and may not need to be heavily weighted in the pricing model.

7. Limitations and Future Improvements

The model has the following limitations that should be considered:

The dataset is US-based and may not fully represent South African medical aid patterns and costs

The model explains 78.35% of the variance, meaning 21.65% of the variance in charges is still unexplained

Linear Regression assumes a linear relationship between features and charges, which may not always hold true To further improve the model the following is recommended

-Collect South African-specific data for better localisation of the model

- Explore polynomial regression to capture non-linear relationships in the data

-Add interaction terms between smoker and BMI, as these two features combined have a strong effect on charges

- Tune Elastic Net hyperparameters for potentially better performance

- Consider using more advanced models such as Random Forest or Gradient Boosting for improved accuracy.

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